

Figure 3.18 Our graphical representations of each team now include a small PNG national flag, downloaded from Wikipedia and loaded using an SVG `<image>` element.

```
d3.selectAll("g.overallG").insert("image", "text")
  .attr("xlink:href", function(d) {
      return "images/" + d.team + ".png";
  })
  .attr("width", "45px").attr("height", "20px").attr("x", "-22")
  .attr("y", "-10");
```

To make the images show up successfully, use `insert()` instead of `append()` because that gives us the capacity to tell D3 to insert the images before the text elements. This keeps the labels from being drawn behind the newly added images. Because each image name is the same as the team name of each data point, we can use an inline function to point to that value, combined with strings for the directory and file extension. We also need to define the height and width of the images because SVG images, by default, have no setting for height and width and won't display until these are set. We also need to manually center SVG images—here the `x` and `y` attributes are set to a negative value of one-half the respective height and width, which centers the images in their respective circles, as shown in figure 3.18.

You can tie image resizing to the button events, but raster images don't resize particularly well, and so you'll want to use them at fixed sizes.

> **Infoviz term: chartjunk**
>
> Now that you're learning how to add images and icons to everything, let's remember that just because you *can* do something doesn't mean you *should*. When building information visualization, the key aesthetic principle is to avoid cluttering your charts and interfaces with distracting and useless "chartjunk" like unnecessary icons, decoration, or skeuomorphic paneling. Remember, simplicity is force.
>
> The term *chartjunk* comes from Tufte, and in general refers to the kind of generic and useless clip art that typifies PowerPoint presentations. Although icons and images are useful and powerful in many situations, and thus shouldn't be avoided just to maintain an austere appearance, you should always make sure that your graphical representations of data are as uncluttered as you can make them.

3.3.2 HTML fragments

We've created traditional DOM elements in this chapter using D3 data-binding for our buttons. If you want to, you can use the D3 pattern of selecting and appending to create complex HTML objects, such as forms and tables, on the fly. But HTML has better authoring tools, and you'll likely be working with designers and other developers who